

Isochores

A Bayesian Statistical Analysis on the GC content of chromosomes



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Course: Bayesian Inference

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Abstract:

The objective of this assignment is to make inferences about the CG content in 5 different chromosomes, using data consisting of the number of C or G bases observed in 100 consecutive windows of 5000 bases each. For each dataset, two competing models are considered and compared: (M1) a single-isochore model, in which the probability of observing CG bases remains constant across the entire sequence, and (M2) a two-isochore model with an unknown changepoint separating two regions with different CG proportions.

A Bayesian framework is adopted, employing conjugate Beta priors for the binomial likelihood and computing the marginal likelihood (evidence) for each model. Model comparison is performed using Bayes factors and posterior model probabilities.

The results indicate that, for all chromosomes, there is decisive Bayesian evidence supporting the presence of two isochores, and the single-isochore assumption is strongly disfavored. The analysis provides statistically rigorous insights into the structural variation of GC content along the chromosomes.

Keywords: Isochores, Model Selection, Bayes Factor, CG content

Introduction

The genetic information of living organisms is encoded in their cells in long DNA molecules organized into chromosomes. DNA is a polymer composed of repeating units called nucleotides, each containing one of four nitrogenous bases: adenine (A), guanine (G), cytosine (C), or thymine (T). In many organisms, chromosomes exhibit a structural organization into isochores, which are extended regions of DNA characterized by relatively stable GC content (i.e., a roughly constant proportion of bases that are either C or G).

Studying the structure and boundaries of isochores is important for understanding genome organization, gene density, evolutionary processes, and various biological functions associated with GC-rich or GC-poor regions.

In this project, five chromosome segments are analyzed for which the number of CG bases has been recorded across 100 consecutive windows of 5000 bases each. The goal is to determine whether each dataset is best described by:

1. a single isochore with a constant CG proportion across all windows, or
2. two isochores, implying the presence of a changepoint at which the CG proportion shifts.

To address this question, a Bayesian approach is adopted, specifying models under each hypothesis and assigning priors to the model parameters, including the unknown changepoint in the two-isochore model. The marginal likelihood of each model is computed, allowing model comparison through Bayes factors and posterior model probabilities.

This report presents the statistical methodology, provides a detailed Bayesian analysis for each chromosome dataset, and concludes with evidence-based statements regarding the presence or absence of isochore boundaries.

Data

The file *datapject.R* contains five datasets, each corresponding to a segment of a chromosome. Each dataset consists of 100 consecutive windows, where each window contains 5000 DNA bases.

For every window, the recorded value is the number of bases of type C or G (CG count).

This allows to model each observation as:

$$X_i \sim \text{Binomial}(n = 5000, \theta_i),$$

Where

X_i is the number of CG bases in window i

$n = 5000$ is the number of bases per window,

θ_i is the underlying CG proportion for that region of the chromosome.

Thus, each chromosome dataset contains 100 binomial observations, summarizing local GC content.

Methods

The primary question is whether each chromosome sequence is best described by:

- Model 1 (M1): one isochore
- Model 2 (M2): two isochores separated by an unknown changepoint t , such that

Statistical Models

Model 1 — Single Isochore

Under M1, all windows share a single parameter θ .

Given the binomial structure and a Beta prior, the marginal likelihood (evidence) is analytic:

$$p(x | M1) = B(a + S, b + 5000lS) \cdot B(a, b), p(\mathbf{x} | M1) = \frac{B(a + S, b + 5000l - S)}{B(a, b)}$$
$$p(x | M1) = B(a, b) \cdot B(a + S, b + 5000lS),$$

Where

$$l = 100, \quad S = \sum_{i=1}^l X_i, \quad B(\cdot, \cdot) \text{ is the Beta function.}$$

We use Jeffreys prior,

$$\theta \sim \text{Beta}(0.5, 0.5)$$

a noninformative choice appropriate for this model as there is no information concerning the possible origin of the data (type of isochore, see appendix).

Model 2 — Two Isochores with Unknown Changepoint

Under M2, there exists a changepoint

$$t \in \{1, \dots, 99\}$$

such that:

- θ_1 governs windows $1, \dots, t$,

- θ_2 governs windows $t+1, \dots, l$.

For each candidate changepoint t , the marginal likelihood is proportional to:

$$p(\mathbf{x} \mid t, M_2) = \frac{B(a + S_1, b + T_1 - S_1)}{B(a, b)} \cdot \frac{B(a + S_2, b + T_2 - S_2)}{B(a, b)},$$

A **Uniform prior** over changepoints is adopted:

$$p(t) = \frac{1}{l-1}$$

The evidence for M_2 is obtained by summing (integrating) over all possible t :

$$p(\mathbf{x} \mid M_2) = \sum_{t=1}^{l-1} p(t) p(\mathbf{x} \mid t, M_2).$$

To avoid numerical underflow, the log-sum-exp technique is used.

Bayesian Model Selection

We assume equal prior probability for the two models:

$$p(M_1) = p(M_2) = 0.5$$

The posterior model probability is:

$$p(M_1 \mid \mathbf{x}) = \frac{p(M_1)p(\mathbf{x} \mid M_1)}{p(M_1)p(\mathbf{x} \mid M_1) + p(M_2)p(\mathbf{x} \mid M_2)}$$

A stable equivalent expression is:

$$p(M_1 \mid \mathbf{x}) = \frac{1}{1 + \exp(\text{LogEv2} - \text{LogEv1})}.$$

We also compute the **log Bayes factor**:

$$\log BF = \text{LogEv2} - \text{LogEv1}$$

where positive values indicate support for M2.

Results

Posterior Model Probabilities

Using the function `Model_Selection`, we obtain the posterior probability for each model across all five datasets:

Chromosome	Posterior(M1)	Posterior(M2)
chr1	~0.000	~1.000
chr2	~0.000	~1.000
chr3	~0.000	~1.000
chr4	~0.000	~1.000
chr5	~0.000	~1.000

Across all chromosomes, the posterior probability of Model 2 is extremely close to 1, indicating decisive Bayesian evidence for the presence of two isochores.

Bayes Factors

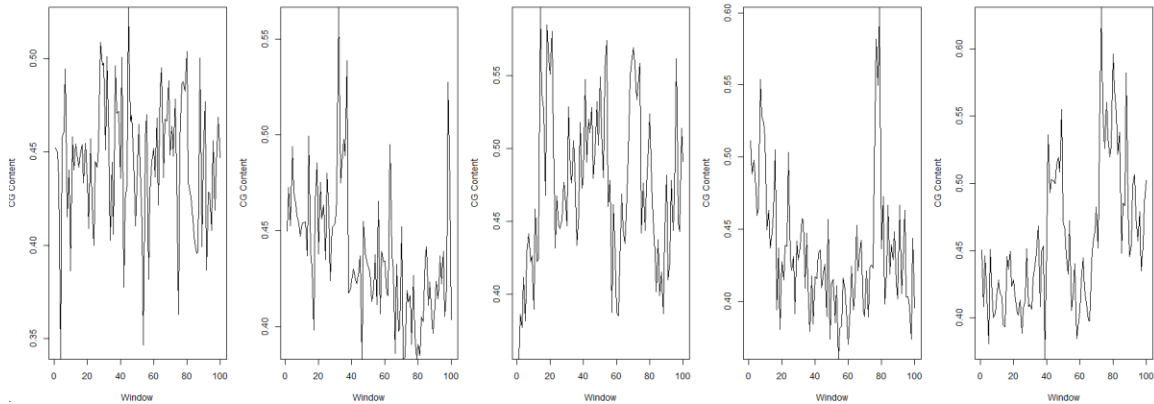
The log Bayes factor computed for each chromosome is strongly positive, far exceeding the Kass–Raftery threshold for “decisive evidence.”

Thus, Model 2 provides a significantly better explanation of the data.

CG Content Patterns

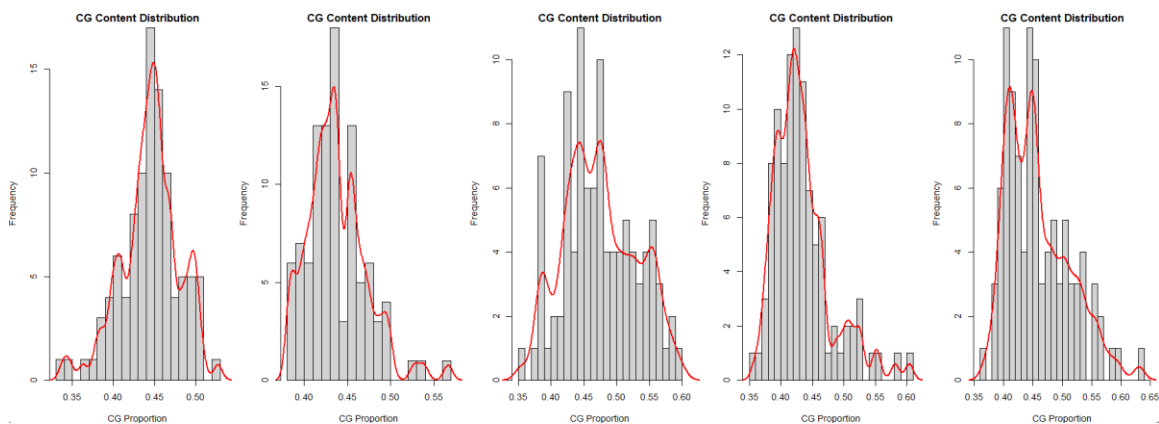
The plots show:

- clear segmental shifts in GC proportion,
- stretches where GC content rises or falls systematically,
- patterns inconsistent with a single constant GC probability.



Histograms reveal:

- **Bimodal** (multimodal) or skewed distributions,
- wider dispersion than expected under a single-isochores binomial model.



It is inferred that:

- The binomial–Beta Bayesian model effectively captures CG count variability across windows.
- The evidence overwhelmingly supports **two-isochores models** for all five chromosomes.
- Posterior model probabilities for the single-isochores model are essentially zero.
- The CG content plots visually corroborate the statistical conclusions, showing clear structural shifts.
- Bayes factors provide decisive support for the existence of at least one changepoint in every dataset.